Economic Analysis of Text and other Non-Standard Data
ISEG Summer School 2026

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12. Ethical Considerations

Now, we move from “What is?” to “What ought to be?”

- Most of the content discussed so far concerns positive questions
  - How does a given model work
  - What are we measuring, etc.
- Today, we will discuss the ethics of text-, image- and audio-data related questions
  - How should models be built from a social viewpoint
  - How should we (not) use them, etc.
  - How should textual, visual or audio data (not) represent our society
- Ethics discussions are necessarily influenced by moral considerations
  - What is good or bad? Just or unjust

Scientists do not operate outside of society

- Positive science is interested in discovery and facts (the "truth" about the world)
- However, positive science interacts with normative considerations in many ways, for example:
  - Which questions do we deem important to research
  - What impact will our discovery have on society
  - What cost are we willing to carry for discovery (not just monetary)
  - What responsibilities do we (not) have as scientists
  - Further examples
- Values and other context-specific characteristics interact with the entire social science pipeline:
  - Research question $\rightarrow$ (data collection) $\rightarrow$ theoretical or empirical analysis $\rightarrow$ interpretation

Let us discuss a historic case study: Census data

- A government wants to establish a comprehensive population registration system for administrative and statistical purposes
- The system should simplify administration and offer perspectives for social research
- Questions:
  - Do you have any ethical concerns
  - Would you be more/less concerned if it is a company/researchers/another actor constructing the comprehensive database? Why?

History has seen horrible examples of unanticipated secondary use

- During the Second World War: Dutch census data used to facilitate the genocide against Jews and others:
  - Data collection during peaceful times
  - Trusted by citizens
  - Later misuse by the Nazis
  - Extremely high death rate of Dutch Jews: 73%
- More: Seltzer, W., & Anderson, M. (2001). The Dark Side of Numbers: The Role of Population Data Systems in Human Rights Abuses. Social Research, 481-513

Ethical Challenges of Generative AI (I/II)

- **Authorship Attribution, Intellectual Property, Copyright, Authenticity, Academic Integrity**
  Plagiarism risks, unclear authorship, ownership of AI-generated content, copyright infringement, and disclosure of AI involvement.
- **Privacy, Trust, Safety**
  Data protection, informed consent, privacy breaches, prompt injection risks.
- **Bias**
  Amplification of social and demographic biases.
- **Misinformation and Deepfakes**
  Generation of realistic false content that undermines trust, democratic discourse, and personal reputation.

Source: Al-kfairy et al. (2024), *Ethical Challenges and Solutions of Generative AI: An Interdisciplinary Perspective*, Informatics.

Ethical Challenges of Generative AI (II/II)

- **Transparency and Accountability**
  Algorithmic opacity, lack of explainability, unclear responsibility for AI-driven harms.
- **Educational Ethics**
  Overreliance on AI, cheating, weakened critical thinking, and inequities in access to AI tools.
- **Social and Economic Impact**
  Job displacement, inequality, concentration of power, and broader societal disruption.

Source: Al-kfairy et al. (2024), *Ethical Challenges and Solutions of Generative AI: An Interdisciplinary Perspective*, Informatics.

Let us discuss further case studies

- Surveillance technology
  - Should governments use face recognition technology? If so, for which applications?
  - How does this question relate to economics research?
- Representation of different groups (e.g., by ethnicity, gender, age)
  - From a normative viewpoint, what is the "right" amount and kind of representation of different societal groups in the media
  - How does this question relate to economics research?
- Accents and labor market outcomes
  - (Under what circumstances) should employers be allowed to consider a worker's accent in their hiring decision
  - How does this question relate to economics research?

How Much Energy Do AI Queries Use?

- Generative AI requires substantial computing resources.
- Recent estimates suggest:

Task	Approximate Energy Use
Google search	$\sim$0.03–0.3 Wh
ChatGPT prompt	$\sim$0.3 Wh
Generating a long answer	$\sim$1–3 Wh
Training a frontier LLM	tens of GWh

- Energy depends strongly on:
  - model size
  - number of generated tokens
  - hardware efficiency
  - batching of requests

Environmental Impact of Large Language Models

- LLMs require significant energy at several stages:
  - **Training** Training frontier models can require tens of gigawatt-hours of electricity.
  - **Inference** Each prompt requires GPU computation and large data center infrastructure.
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- Environmental impacts include:
  - electricity demand
  - carbon emissions
  - water consumption for cooling
  - hardware manufacturing and e-waste

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- **Policy**
  - Increase transparency in AI energy use
  - Develop standards for reporting AI carbon footprints